

Step 1: Start Hadoop

In Ubuntu terminal:

```
start-dfs.sh  
start-yarn.sh  
jps
```

You should see:

```
NameNode  
DataNode  
ResourceManager  
NodeManager  
SecondaryNameNode
```

Step 2: Create Input File

Create sample data:

```
nano input.txt
```

Add:

```
hello hadoop hello world  
python with hadoop  
hello python
```

Save:

CTRL + X → Y → Enter

Step 3: Create Mapper Python File

Create mapper:

```
nano mapper.py
```

Paste:

```
#!/usr/bin/env python3  
import sys
```

```
for line in sys.stdin:
    line = line.strip()
    words = line.split()

    for word in words:
        print(f"{word}\t1")
```

Save.

Make executable:

```
chmod +x mapper.py
```

Step 4: Create Reducer Python File

Create reducer:

```
nano reducer.py
```

Paste:

```
#!/usr/bin/env python3
import sys

current_word = None
current_count = 0

for line in sys.stdin:
    word, count = line.strip().split('\t')
    count = int(count)

    if current_word == word:
        current_count += count
    else:
        if current_word:
            print(f"{current_word}\t{current_count}")
            current_word = word
            current_count = count

if current_word:
    print(f"{current_word}\t{current_count}")
```

Save.

Make executable:

```
chmod +x reducer.py
```

Step 5: Put File into HDFS

Create input directory:

```
hdfs dfs -mkdir /input
```

Upload file:

```
hdfs dfs -put input.txt /input
```

Check:

```
hdfs dfs -ls /input
```

Step 6: Run Hadoop Streaming Job

Find streaming jar:

```
find $HADOOP_HOME -name "**streaming*.jar"
```

Example output:

```
/usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar
```

Run:

```
hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-3.3.6.jar -files  
/home/hduser/mapper.py,/home/hduser/reducer.py -mapper "python3 mapper.py" -reducer  
"python3 reducer.py" -input /input/input.txt -output /output2
```

If output already exists:

```
hdfs dfs -rm -r /output
```

Then rerun.

Step 7: View Output

```
hdfs dfs -cat /output/part-00000
```

Expected:

hadoop 2
hello 3
python 2
with 1
world 1

Folder Structure

mapper.py
reducer.py
input.txt

Viva Explanation